

RegUNeXt with Hybrid Attention for Advanced Segmentation for tumorous Region in Brain Tumor.

Abstract—The segmentation of brain tumor plays a significant role in identification of the tumorous region, thus facilitating the formulation of effective treatment strategies to overcome the high mortality rate. Manual segmentation of brain tumors frequently results in delays in diagnosis and highlights the necessity for implementing automated solutions. In the proposed segmentation framework a novel approach has been employed that uses pre-trained RegNeXt blocks as a backbone of a U-Net-based segmentation model. The segmentation framework has been further refined through the incorporation of Spatial Attention Modules (SAM) and Channel Attention Modules (CAM) for enhanced feature refinement. To optimize the performance, the focal loss has been utilized to address class imbalance and the Adam optimizer for robust optimization. The framework was evaluated using the BRATS 2020 T1-weighted MRI dataset, resulting in a Dice Score Coefficient (DSC) of 93.89%. This method highlights the combination of attention-augmented architectures with high-capacity feature extractors to achieve accurate and efficient segmentation of brain tumors.

Index Terms—Brain Tumor, Backbone, Channel Attention Module (CAM), Dice Score Coefficient (DSC), RegNext, Spatial Attention Module (SAM), U-Net

I. INTRODUCTION

Brain tumors are the abnormal growth of cells in the brain, which are serious and life-threatening. Early identification and localization of brain tumors are crucial for efficient treatment to decrease the mortality rate in a patient [2]. Magnetic Resonance Imaging(MRI) is widely used for brain tumor evaluation as it is non-invasive to detect the tumor. Different MRI modalities such as T1, T2, T1CE, and FLAIR offer additional information, capturing structural and pathological changes in brain tissues [1]. These techniques play a key role in the accurate analysis and diagnosis of brain tumors in clinical practice.

The domain of medical imaging analysis has undergone significant transformation due to recent advancements in computational techniques [3]. Traditional methodologies in Machine Learning (ML) require handcrafted features that require extensive domain expertise to design task-specific representations [5]. Although these methods demonstrate effectiveness in certain contexts, their capacity for generalization across diverse datasets is often limited, primarily due to the constraints in feature representation. In contrast, the emergence of Deep Learning (DL) methodologies, especially Convolutional Neural Networks (CNNs), brought a revolutionary change in the field of detection of brain tumor [4]. These advanced

networks possess the ability to learn hierarchical features autonomously from raw data. This inherent capability results in superior performance of DL models over traditional ML approaches, particularly in complex applications such as tumor segmentation, where the intricacies of data representation play a critical role.

A. Related Work

Isaza et al.[2] introduced the CrossTransUnet model, which integrates a cross-attention mechanism within a four-stage deep encoder-decoder framework. This model employs separable convolution layers, effectively lowering computational demands while ensuring robust segmentation capabilities. In [3], a lightweight U-Net model was developed for optimal brain tumor segmentation by selecting the most informative slices. This approach prioritizes the reduction of computational complexity while ensuring efficient tumor localization. Zakariah et al.[4] proposed the Dual Vision Transformer-DSUNET model, which demonstrates a novel combination of dual vision transformers and U-Net, enhanced using feature fusion techniques. The implementation of dual attention mechanisms facilitates the effective capture of both global context and detailed spatial information, improving segmentation of multi-modal MRI data. In [1], a finetuned U-Net architecture is used for glioma detection from the Brain MRI Scan. The development of brain tumor segmentation by developing an enhanced model founded on a modified U-Net architecture in [5] aims at being precise and robust. Structural refinements were made to the modified U-Net, which included adjustments to kernel size, the number of channels, dropout ratios, and a shift from the ReLU activation function to Leaky ReLU.

B. Novel Contribution

The main contributions of the proposed frameworks are as follows:

- 1) A robust U-Net-based framework uses pre-trained CNN blocks along with a Hybrid Attention Module (HAM) that incorporates SAM+CAM to enhance the segmentation of brain tumors in T1-weighted MRI images. This architecture helps to capture fine details and complex tumor boundaries.
- 2) Incorporation of pre-trained CNN blocks in the encoder part addresses the vanishing gradients problem, allowing the network to pull out both lower-level and higher-level

features more effectively, which leads to better feature sharing and a more stable model.

- 3) Integration of SAM and CAM in the encoder-decoder structure improves the feature maps by highlighting important areas in spatially relevant regions in SAM and key feature channels in CAM, resulting in improved segmentation accuracy and robustness.

The rest of the paper contains the proposed architecture in Section III. The methodology in Section II contains the pre-processing of the dataset and the proposed framework. The results obtained from the proposed model and discussion are contained in Section IV. The conclusion and future scope have been provided in Section V

II. METHODOLOGY

A. Dataset

The BraTS 2020 dataset has been utilized to segment brain tumors, consisting of 369 MRI scans containing different imaging modalities. Each scan includes five Neuroimaging Informatics Technology Initiative(NIfTI) files that provide ground truth for the tumor and four MRI modalities, i.e, T1-weighted, T1-weighted with contrast, T2-weighted, and T2-weighted fluid-attenuated inversion recovery (FLAIR), as outlined in the table I. This study utilizes 100 T1-weighted NIfTI scans, each scan containing 155 slices. The proposed framework is designed for the set of T1-weighted images. Each slice has a resolution of 240×240 pixels, making it suitable for single-modality segmentation. The brain tumor slices have been illustrated in Fig. 1

TABLE I: Details of Brats2020 Dataset.

Dataset	Brats2020
Data format	3D NIfTI format (. nii.gz files)
Image modalities	T1, T2, T1ce and Flair
Training data of T1 weighted	100 patients
No. of T1 slices per patient	155 slices
No. of mask slices per patient	155 slices
Image size	240x240

B. Pre-processing

In this study, a subset of the BRaTS 2020 dataset has been utilized, which includes T1-weighted brain MRI scans from 100 patients. Each of these 3D MRI scans are sliced into 155 2D images, resulting in a total of 15,500 slices. To ensure the intensity levels across all samples, the 2D samples were normalized.

The utilization of a smaller brain tumor dataset has been allowed for efficient model optimization and reduced computational overhead during training. MRI can often show various intensity levels due to differences in imaging machines and procedures. Because of these variations in intensity, the deep learning models are computationally demanding without proper normalization. To address this, MRI scans were intensity normalized by scaling pixel values to a range(0 to 1), ensuring consistency throughout the dataset. Finally, the dataset has been split into training, validation, and testing sets in an 80:15:5 ratio.

C. Model Architecture

RegNeXt is an improved variant of ResNeXt, designed to improve feature learning, data handling, and computational efficiency. It uses grouped convolutions focusing on cardinality to learn features efficiently without significantly increasing computational cost. Additionally, RegNeXt includes residual connections to keep the gradient flow and adds regularization techniques like dropout, stochastic depth, and batch normalization to mitigate overfitting. The structure of the RegNext backbone is shown in Fig.2.

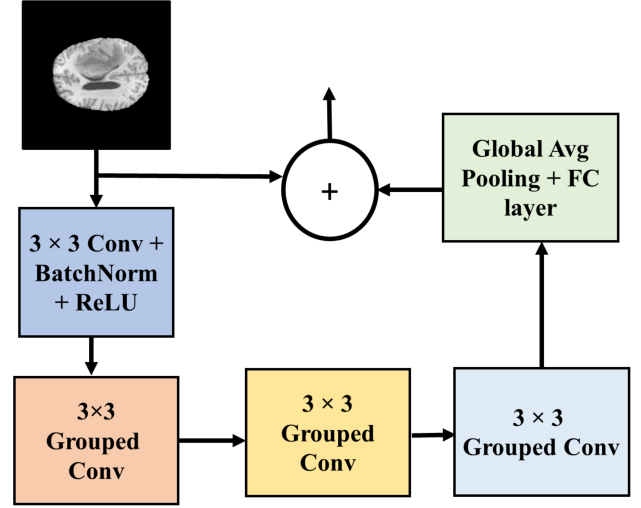


Fig. 2: RegNeXt Backbone Structure

The Spatial Attention Module (SAM) is an effective mechanism that enhances feature representation by focusing on critical regions within a slice. Unlike channel attention, which focuses on feature relationships across various channels. SAM generates an attention map that highlights spatially important areas. It achieves this by including both max and average pooling to recognize the most relevant regions for improved feature refinement.

The Channel Attention Module(CAM) enhances feature representation by prioritizing feature channels, unlike SAM, which highlights critical spatial regions. CAM assigns different levels of importance to different feature channels. It achieves this by utilizing global average pooling and max pooling to identify the most informative channels, allowing the model to focus on key features and improve segmentation accuracy.

U-Net is a widely used deep-learning framework for medical image segmentation that features an encoder-decoder architecture with skip connections that transpose each other. The encoder extracts key features through convolutional and pooling layers, while the decoder progressively upsamples

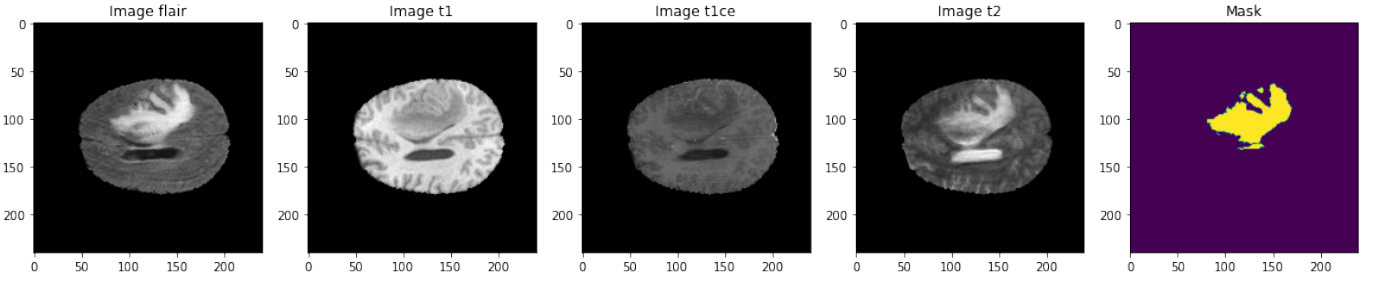


Fig. 1: Slices from Brats2020 with Different Modalities and its Mask

these features to generate precise segmentation maps. The skip connections bridge between corresponding encoder and decoder layers (with the same spatial dimension), preserving low-level spatial details and improving segmentation accuracy.

D. Focal Loss

The performance of brain tumor segmentation is heavily influenced by the class balance ratio. In brain tumor segmentation, some tumors are significantly harder to detect compared to larger ones due to their spatial properties. This imbalance can lead to poor model performance, where the model favors larger regions and may ignore smaller regions. To address this issue, focal loss is used, which is an improved version of cross-entropy loss. Focal loss reduces the influence of well-classified features while giving more importance to hard-to-classify regions, improving segmentation performance where the model focuses more on difficult cases, such as small tumor areas. The formula for Focal Loss is given in Equation(1) :

$$F = \alpha \cdot (1 - p_t)^\gamma \cdot -\frac{1}{N} \sum_{i=1}^N [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)] \quad (1)$$

Where: α is a balancing factor for class imbalance, p_t is the predicted probability for the true class, γ is the focusing parameter that down-weights easy features, and BCE Loss is the binary cross-entropy loss.

E. Evaluation Metrics

To evaluate the effectiveness of the segmentation model, the proposed framework relies on several key metrics, including the Dice coefficient(DSC), Intersection over Union, accuracy, and precision. These metrics measure the efficiency and performance of the model in case of pixels in the target region and aligns with the ground truth. Specifically, these measure how accurately the model segments the correct tumor region, minimizing false positives and false negatives.

$$DSC = 2 \cdot \frac{|A_{pred} \cap A_{gt}|}{|A_{pred}| + |A_{gt}|} \quad (2)$$

In Equation(2), A_{pred} represents the area predicted by the model, while A_{gt} is the actual ground truth mask.

For IoU, it shows how much the predicted mask overlaps with the ground truth mask for the predicted regions, divided by the total area of both masks.

$$IoU_i = \frac{TP_i}{TP_i + FP_i + FN_i} \quad (3)$$

In Equation(3), it calculates the average of all the IoUs for $i = 1$ to the number of pixels in the object(the tumor region of the brain).

Accuracy is expressed as in Equation(4).

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (4)$$

$$Precision = \frac{TP}{TP + FP} \quad (5)$$

Precision is expressed as in Equation(5)

III. PROPOSED ARCHITECTURE

The proposed framework has been built on a U-Net framework by adding both a SAM and CAM, using a pre-trained RegNeXt as the backbone. The RegNeXt backbone, which comes with pre-trained weights and biases, boosts feature extraction and enhances the overall performance of the segmentation procedure. This proposed segmentation framework efficiently captures dependencies in spatial and channel-dependent features. The added hybrid attention modules refines the important feature representations for greater efficiency in segmentation. The framework has been trained on single-channel T1 images from the BraTS2020 dataset, where the RegNeXt-based encoder pulls out hierarchical features. SAM dynamically enhances spatial details by focusing on important areas, and CAM emphasizes the most relevant feature channels, highlighting essential features as well as reducing redundancy. The skip connections bring low-level spatial details into the decoder, which then creates the segmentation mask through upsampling and feature integration.

Finally, a 1×1 convolution generates the binary segmentation mask, and during training, focal loss is used to tackle class imbalance and focus on segment areas. This setup guarantees precise outlining of tumor structures. Additionally, data augmentation is used to improve the model and is trained with optimizers like Adam and RMSprop to refine boundary definitions. The strong feature extraction abilities of RegNeXt, along with the advantages of SAM and CAM, enable the

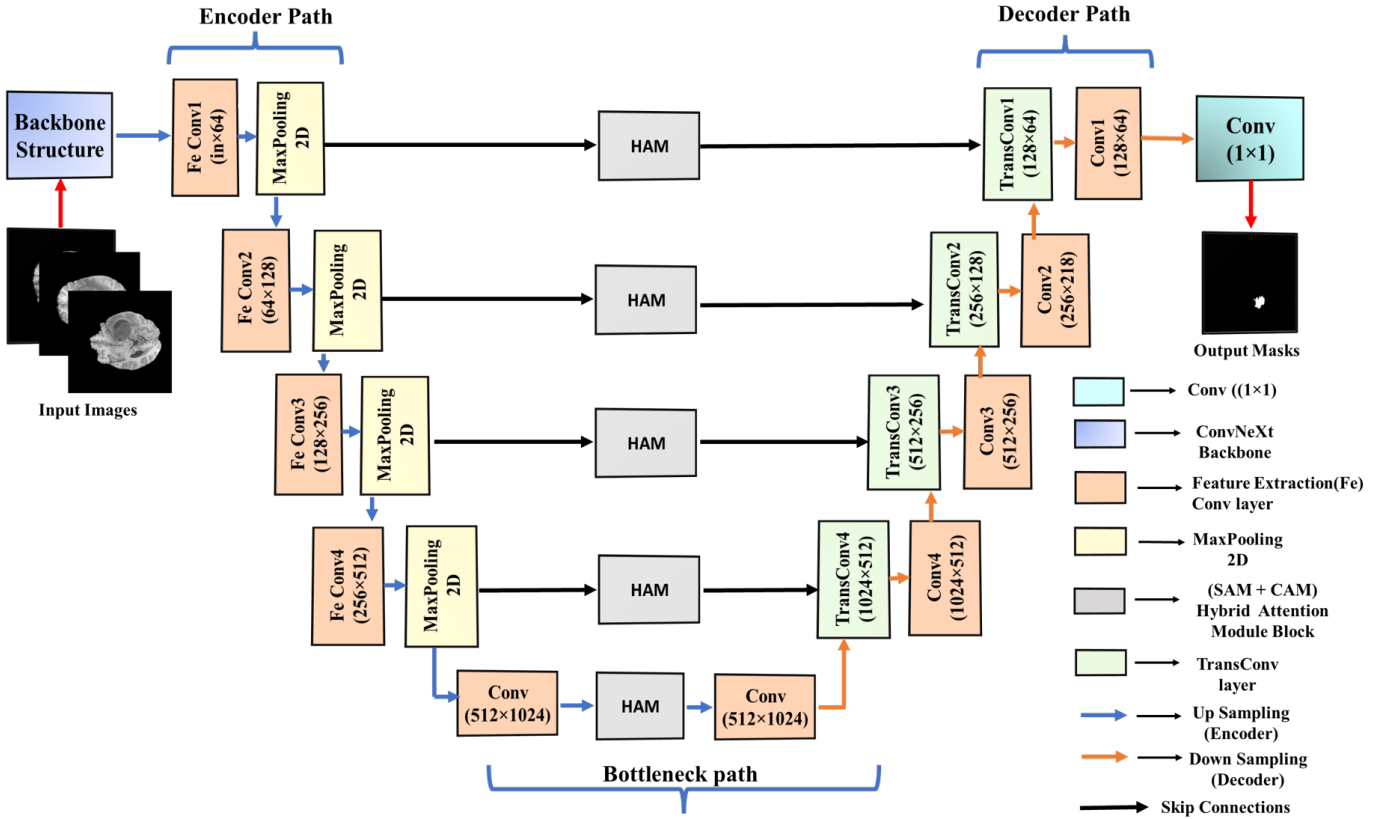


Fig. 3: Proposed novel attention-based framework for Brain tumor segmentation

proposed framework to capture both global and local features effectively, leading to accurate segmentation of complex tumor structures. The proposed segmentation framework along with the hybrid attention module has been represented in Fig. 3.

IV. RESULTS AND ANALYSIS

A. Implementation Details

The proposed framework has been built using Python along with the PyTorch library. The framework has been experimented with Adam and RMSprop optimizers. Adam improves the performance by adjusting learning rates by combining momentum with RMS, making it adaptable for many tasks. On the other hand, RMSprop helps keep training stable by adjusting learning rates based on gradients, especially useful for smaller datasets.

During training, we set the learning rate to 0.001 and trained the model for 45 epochs with a batch size of 16, considering our limited computing resources. The details of the software setup are provided in Table II.

B. Analysis of results

1) *Results:* The results have been featured in Table III, where the model evaluates the performance of the segmentation framework with and without SAM and CAM. The metrics, such as DSC and IOU, demonstrate

TABLE II: Software specifications.

Software specifications	
Platform	Google Colab
Deep Learning Frame work	PyTorch
Programming language	Python 3.10.12
GPU	T4 GPU
CPU	Intel(R) Xeon(R) CPU @ 2.20GHz
RAM	12.6GB

that the hybrid attention module significantly improves the segmentation performance in comparison with the model without the attention module, thereby highlighting its effectiveness in the segmentation. RegNeXt-based segmentation architecture efficiently captures essential tumor features while aligning with the ground truth. SAM improves spatial feature extraction by focusing on important tumor regions, while CAM ensures the performance of the framework on the most important feature channels, further boosting segmentation accuracy.

The hybrid attention module, along with the RegUNeXt, addresses the challenge of class imbalance, which arises due to non-tumorous pixels. This improvement makes the framework more robust and improved its ability to accurately segment tumors. The segmented outputs across various brain slices with

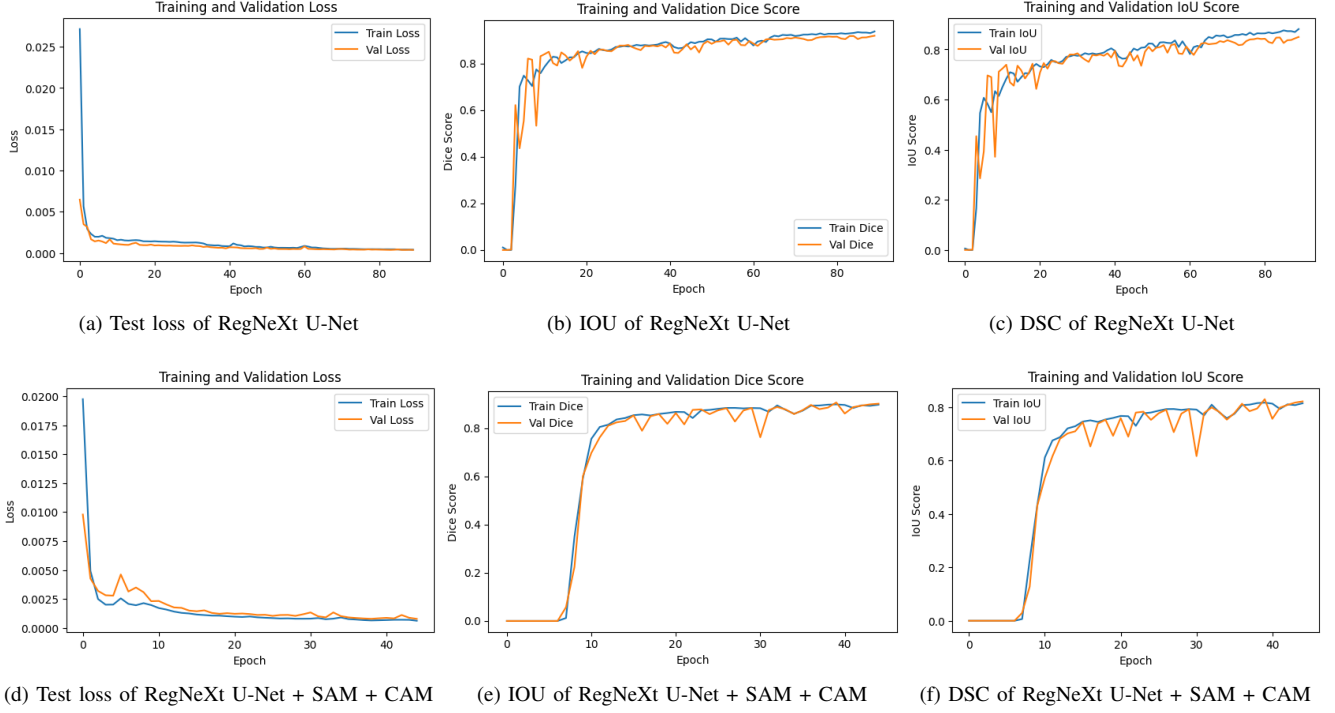


Fig. 4: Test loss, DSC, and IoU graphs of training and validation results

TABLE III: Evaluation of the Proposed Segmentation Framework.

Model	Loss Function	Optimizer	Loss	Dice Coefficient(%)	IoU (%)	Accuracy(%)	Precision(%)
U-Net	Focal	Adam	3×10^{-3}	87.62	78.53	97.66	95.41
		RMS Prop	22×10^{-3}	72.45	63.45	92.13	88.73
RegNext U-Net	Focal	Adam	7×10^{-5}	90.29	82.30	99.55	95.37
		RMS Prop	16×10^{-5}	88.65	79.62	99.02	87.38
RegNeXt U-Net + SAM + CAM	Focal	Adam	4×10^{-4}	93.89	88.48	99.66	93.58
		RMS Prop	12×10^{-4}	92.69	86.37	99.70	95.71

and without the hybrid attention module have been shown in Fig. 5 and 6, respectively.

2) *Discussion*: The RegUNeXt framework has scored a DSC of 90.29% and demonstrated an IoU of 82.30%, which indicates a high feature extraction ability but is still limited. With the incorporation of SAM and CAM, the performance has been significantly improved, with a Dice of 93.89% and an IoU of 88.48% using the RegUNet+SAM+Cam framework for segmentation. The comparative results highlight the effectiveness of the SAM and CAM in improving the accuracy. By focusing on spatially significant tumor regions and capturing fine-grained tumor regions, SAM enhances the ability of the model to distinguish between tumor and non-tumor regions.

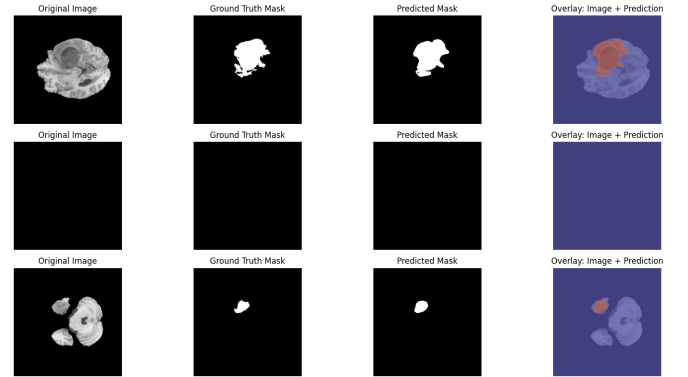


Fig. 5: Visualization of Results without SAM + CAM

The impact of SAM and CAM improves performance is evident in the evaluation metrics. Without the usage of the hybrid attention module along with RegUNeXt, it achieved a precision of 95.37%, an accuracy of 99.55%. The hybrid attention module increases the efficiency with a percent of 3-6%. The performance parameters achieved are 93.89% for

precision and 99.66% for accuracy. This demonstrates that the strength of hybrid attention module enhances the feature representation while effectively reducing the false positives in tumor regions, leading to more accurate segmentation.

3) *State-of-the-art Comparison*: The study introduces a hybrid attention-based framework for brain tumor segmentation

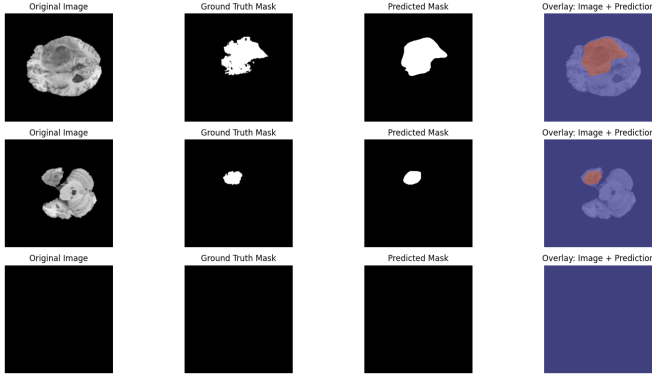


Fig. 6: Visualization of Results with SAM + CAM

utilizing T1-weighted MRI scans from the Brats 2020 dataset. This attention-based mechanism significantly improves segmentation accuracy. The proposed model achieves a dice score of 93.89%, and has outperformed the state-of-the-art segmentation models. The state-of-the-art comparison of different segmentation frameworks using the BraTS-2020 dataset has been mentioned in Table IV

TABLE IV: State-of-the-art(SOTA) of Different Segmentation Framework

Author	Method Used	DSC (%)
Ali et al. (2022)[1]	U-Net	90.7
Anaya-Isaza et al. (2023) [2]	CrossTransU-Net	93.0
Hernandez-Gutierrez et al. (2024)[3]	Light Weight U-Net	86.0
Zakariah et al. (2024) [4]	Dual Vision Transformer-DSU-Net	91.47
Alquran et al. (2024) [5]	U-Net	90.2
Proposed Framework (2025)	RegNeXt U-Net + SAM + CAM	93.89

V. CONCLUSION

This proposed framework provides an in-depth analysis of the effects and benefits of the SAM and CAM on a small subset of the BraTS 2020 dataset. The utilization of RegNeXt as the backbone of U-Net significantly enhances the performance of the semantic segmentation of the brain tumor with respect to the ground truth mask. The pretrained CNN-based model leverages weights and biases, offering a well-initialized starting point for the CNN blocks of U-Net. This modification to the base U-Net segmentation architecture leads to improved identification of tumorous and non-tumorous regions. The novel framework incorporates SAM+CAM, where SAM refines spatial features and addresses class imbalance in pixel-wise classification. The CAM enhances feature selection by prioritizing the most relevant channels, ensuring that crucial tumor-related features are preserved. This dual attention mechanism strengthens feature representation and segmentation accuracy. The proposed framework plays a vital role in automatically diagnosing brain tumors at different stages, improving clinical

decision-making.

This proposed framework effectively handles the class imbalance issue and uses the pretrained weights of RegNeXt to efficiently extract the tumorous features. The primary objective of this segmentation framework is to enhance performance, which has been successfully achieved, yielding a Dice Similarity Coefficient (DSC) of 93.89%. In the future, the architecture can be further improved and adapted using different CNN models to achieve even better segmentation performance, particularly in distinguishing different tumor classes from non-tumorous regions.

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